

When Does Population-Mobility OD Injection Help Traffic Forecasting? A Capacity-Gap Account Across Eleven Models

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Abstract—Road traffic is a downstream consequence of population movement, so a forecast of origin–destination (OD) flow, routed onto the road network, should tell a traffic model something its own history does not. We test when it does, on Seoul (500 OD zones; 121 volume and 396 speed sensors; 2023–2026) under 30-fold rolling-origin validation across eleven forecasting models of 31k–774k parameters. Injected through a learned per-node gate, routed OD reduces a lightweight graph-recurrent model’s MSE by 7.2% (volume) and 8.2% (speed); the road-network routing is essential, and most of the gain is calendar-shaped, with a real day-specific share. The same mechanism does not generalize: among ten published architectures it helps only the two smallest and is significantly harmful, on at least one task, for every architecture above 10^5 parameters – an ordering by model size (Spearman $\rho = 0.89$ on volume) that scaling a single architecture reproduces, and that an oracle experiment shows is not caused by forecast error. A fusion-free, ensemble-controlled test confirms this *capacity gap*: using the OD-injected model as an ensemble partner beats using a second random seed for every model up to 62k parameters, on both tasks, but never for any model of 95k parameters or more ($\rho = 0.77$, $p = 2.5 \times 10^{-5}$). Validation-gated per-sensor selection between the two checkpoints, a seemingly natural remedy, turns out to be dominated by simple averaging. For practice, auxiliary mobility signals are worth injecting into compact models; for large models, the same compute is better spent on a second seed.

Index Terms—Traffic prediction, origin–destination flow, road network routing, spatio-temporal knowledge injection, urban computing, empirical study, graph neural networks.



1 INTRODUCTION

Short-term forecasts of road traffic volume and speed underpin congestion management, route guidance, and signal control. The dominant modeling paradigm treats this as a purely *endogenous* problem: a deep spatiotemporal model observes past sensor readings on a road-network graph and extrapolates them, using graph convolution [1]–[3], attention [4]–[6], or learned graph structure [7]–[9]. This framing ignores information that is causally *upstream* of traffic: the origin–destination (OD) movement of the population that produces it. A commuter’s decision to travel precedes, by minutes to hours, the vehicle that decision puts on a specific sequence of road links. If population movement can be forecast, and correctly attributed to the links it will cross, it should give a traffic model information its own history does not contain.

The question is practically relevant. Transport authorities increasingly have access to aggregate population-mobility products (mobile-carrier location records, card-transaction geodata) at cadences fast enough to be predictive. Injecting such a signal into an already deployed model is far cheaper than re-architecting around a larger one – *if* it reliably helps. Knowing when it helps, and when it does not, is therefore as valuable to a practitioner as a single positive result.

Prior work that brings regional context into road-level prediction does so through a *learned*, implicit mapping – e.g., [10] attends from region-level embeddings (LTE traces, POIs, imagery) to road-level representations on Seoul speed data. We test an *explicit* channel instead: a road-hierarchy-weighted shortest-path assignment routes the forecast OD flow onto the links it would physically cross, so the signal injected at a sensor has a concrete, checkable justification.

We evaluate this with 30-fold rolling-origin validation (12-month train / 1-month validation / 1-month test, 1-month stride), paired significance tests on every comparison, and two sensor modalities. The study unfolds in three steps, each answering the question the previous one raises.

Step 1 – a lightweight model benefits. Injecting the routed OD forecast through a learned per-node, per-step gate reduces our lightweight graph-recurrent model’s MSE by 7.2% (volume, 29/30 folds) and 8.2% (speed, 30/30), both $p < 10^{-7}$. Controls show the road-network routing is essential (an unrouted signal does nothing) and that much of the gain is calendar-shaped, with a real but smaller share (34%) carried by day-specific OD content.

Step 2 – uniform injection does not generalize. Applied to ten published architectures spanning 31k–774k parameters – STID [11], GMAN [5], STGCN [1], PDFormer [12], STAEformer [6], MTGNN [9], DCRNN [2], GTS [8], Graph WaveNet [3] and AGCRN [7] – the same mechanism significantly helps only the two smallest (STID and GMAN, on volume), and is significantly harmful on at least one task for every architecture above 10^5 parameters. The effect tracks model size (Spearman $\rho = 0.89$ on volume), and scaling a

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- Code, analysis scripts, and result summaries: <https://github.com/SuminHan/traffic-od-injection-capacity-gap>

single architecture (STID) reproduces the same decay and sign reversal. An oracle experiment rules out OD forecast error as the cause. We call this the *capacity gap*: once a model is expressive enough to infer what the signal says, a fixed fusion mechanism cannot extract net value from it.

Step 3 – an ensemble-controlled test confirms the gap.

Uniform injection measures the signal through one fusion mechanism, which might simply fail to extract value the signal carries. We therefore test the injected checkpoint without any fusion or selection mechanism, as an ensemble partner, against the right control: a second plain checkpoint differing only in seed. For every model up to 62k parameters the OD-injected partner is significantly better, on both tasks – even on speed, where uniform injection showed no gain. For no model of 95k parameters or more is it better, and for five combinations it is significantly worse. The capacity gap is thus not an artifact of the fusion mechanism, and it is sharper under this control ($\rho = 0.77$) than under uniform injection ($\rho = 0.67$). Along the way, a validation-gated per-sensor selector – a natural candidate remedy – turns out to be dominated by simply averaging the two checkpoints.

Our contributions are:

- **A capacity-gap account of auxiliary-signal injection**, established across eleven models under three independent tests – uniform injection across architectures, a capacity sweep within a single architecture, and a fusion-free ensemble-controlled comparison – with an oracle experiment and alternative fusion mechanisms ruling out forecast error and fusion expressiveness as explanations.
- **An ensemble-controlled evaluation protocol** for auxiliary signals: comparing the signal-injected model with a second-seed model as an ensemble partner separates the signal’s contribution from generic ensembling, and shows that a seemingly effective per-sensor selection remedy owes most of its gain to the latter.
- **A routed OD signal and a reproducible study design**: road-network routing of an OD forecast into a sensor-level signal, 30-fold rolling-origin evaluation throughout, routing and calendar-confounding controls, and reimplementations-fidelity checks; code and per-fold results are public.

Section 2 reviews related work, Sections 3–5 define the problem, method and setup, Section 6 presents Step 1, Section 7 presents Steps 2 and 3, Section 9 discusses limitations, and Section 10 concludes. The supplementary material contains a fusion-architecture ablation, the per-sensor selection experiments in full, cross-dataset protocol details, two exploratory findings, and the reimplementations-fidelity check.

2 RELATED WORK

2.1 Traffic Forecasting Models

Classical approaches – ARIMA [13] and Kalman filtering [14] – model each sensor as a low-dimensional linear time series and cannot capture the nonlinear spatial propagation of congestion; deep spatiotemporal models have repeatedly been shown to dominate them on road-sensor

benchmarks [2], [3], so we do not re-run them here. The now-dominant paradigm combines graph-structured spatial mixing with temporal sequence modeling. STGCN [1] pairs graph convolution with gated temporal convolution; DCRNN [2] models diffusion on a directed graph inside a recurrent encoder-decoder; ASTGCN and ASTGNN [4], [15] add spatial and temporal attention; STSGCN [16] models local spatiotemporal correlation jointly; T-GCN [17] and ST-ResNet [18] combine graph or convolutional spatial modeling with recurrent or residual temporal modeling. A second family learns the graph itself: Graph WaveNet [3] learns a dense adaptive adjacency alongside dilated causal convolutions, AGCRN [7] learns node-specific parameters and adjacency from node embeddings, GTS [8] samples a discrete structure via Gumbel-softmax, and MTGNN [9] learns a graph for arbitrary multivariate series. Attention-based and efficiency-oriented designs include GMAN [5] (stacked spatial/temporal attention), D2STGNN [19], PDFormer [12] (explicit propagation delay), STAEformer [6] (adaptive embeddings on a vanilla Transformer), STID [11] (identity embeddings with an MLP backbone) and BigST [20]. We use ten of these – STID, GMAN, STGCN, PDFormer, STAEformer, MTGNN, DCRNN, GTS, Graph WaveNet and AGCRN – as the published architectures of Section 7, chosen to span a $25\times$ range of parameter counts and qualitatively different treatments of spatial structure (fixed diffusion, learned adjacency, learned discrete structure, attention, and graph-free embeddings).

2.2 Population/Regional-Knowledge-Aware Traffic Prediction

Prior work has explored injecting exogenous population or regional context into road-level traffic models. Most directly related is [10], discussed above, which embeds real-time region-level knowledge via a *learned* bipartite spatial transform attention to convert region-level to road-level representations, evaluated on Seoul traffic speed and living-population data in three downtown districts. Our approach differs in kind, not just degree: instead of a learned implicit region-to-road mapping, we compute an *explicit* shortest-path assignment over the actual road network, so a given sensor only receives signal from OD pairs whose routed path provably crosses that sensor’s link. Our own prior work [21] established the foundational floating-population embedding this line of work builds on, applied to traffic prediction without explicit road-network routing.

2.3 Knowledge Adaptation Between Data-Rich and Data-Sparse Sources

Beyond traffic specifically, knowledge adaptation from a data-rich source to a data-sparse target is a recurring pattern in urban prediction. Within a single city, [22] transfers demand-forecasting knowledge from a station-intensive public-transit mode (bus) to station-sparse modes (train, light rail, ferry) via a memory-augmented multi-task network with a learned *boost/eliminate* dual-vector gate, reporting that naive feature concatenation of the two sources actively *hurts* accuracy relative to the target-only model – a finding we independently reproduce in a different setting in Appendix A (supplementary material). We adapt this

dual-gate mechanism as one of the injection mechanisms tested in Section 7. Across cities, RegionTrans [23] matches regions between cities to transfer spatiotemporal prediction knowledge from a data-rich source city to a data-scarce target city; CrossTReS [24] selectively re-weights source-city regions by relevance before transfer; and, in a different urban sensing domain, FLORAL [25] adapts semantically related dictionaries across modalities for air-quality prediction. Across this broader literature, reported successes share a consistent structural signature: the target is a genuinely data- or capacity-limited learner – few stations, a short training history, a data-poor city, or, in our terms, a lightweight model. This pattern is consistent with, and directly motivates, the capacity-gap explanation we develop in Section 7.

2.4 Foundation and Universal Models for Spatiotemporal Prediction

A separate and rapidly growing line of work pursues generalization *across* datasets, cities, and even modalities rather than richer per-dataset signal. UniST [26] pre-trains a single Transformer across diverse spatiotemporal domains via masked-token modeling and adapts to new scenarios with knowledge-guided prompts in few-/zero-shot settings. GPT-ST [27] and STEP [28] similarly pursue generative or predictive pre-training for spatiotemporal graphs. Most recently, large language models have been adapted directly to spatiotemporal forecasting: UrbanGPT [29] couples a spatiotemporal dependency encoder with an instruction-tuned LLM for zero-shot urban prediction; ST-LLM [30] tokenizes per-location time steps with spatial-temporal embeddings and partially freezes LLM attention to adapt it to traffic data; and OpenCity [31] combines Transformer and graph-neural-network components into an open foundation model with promising zero-shot and scaling behavior across cities. This direction is orthogonal to ours – these works study generalization of a shared backbone across datasets and cities, whereas we study a single-city, cross-modality knowledge-injection mechanism and characterize precisely when and why it helps, which is a question that remains unanswered regardless of how the underlying backbone is pre-trained or shared.

2.5 Evaluation Rigor in Spatiotemporal Forecasting

Benchmarking studies have shown that reported gains in multivariate time-series and traffic forecasting are often inconsistent across datasets and evaluation settings, and that heterogeneity among datasets explains much of the disagreement [32]. More generally, variance from random initialization and data sampling can be as large as the improvements being claimed, so comparisons need multiple trials and controls for these sources of variation [33]. Averaging independently trained networks is itself a strong and well-known way to improve predictions [34]. Our ensemble-controlled test combines these observations: it asks whether a signal-injected model improves an ensemble more than an equally trained model that differs only in seed, which separates the signal’s contribution from both seed variance and generic ensembling.

3 PRELIMINARIES AND PROBLEM DEFINITION

Road network. The capital-region road network is represented as a directed graph $\mathcal{G} = (\mathcal{V}_r, \mathcal{E}_r)$ with 315,048 nodes and 430,675 directed edges, built from MOCT LINK/NODE shapefiles restricted to the capital-region bounding box. Each edge $e = (u, v)$ has weight

$$w_e = \ell_e \cdot \rho_e^\alpha, \quad (1)$$

where ℓ_e is physical edge length in meters, $\rho_e \in (0, 1]$ is a road-hierarchy discount factor mapping the seven MOCT road-rank categories to values from ≈ 0.50 (local road) up to 1.00 (motorway), and $\alpha = 0.9$ controls the strength of the hierarchy preference relative to raw distance in shortest-path routing (higher α more strongly prefers major roads even at the cost of extra distance).

OD movement. $\mathcal{X}_{od} \in \mathbb{R}^{T \times N \times N}$ is the hourly origin–destination population flow between $N=500$ administrative (dong-level and, outside Seoul proper, gu/si-level) units for T hours spanning 2023-01 through 2026-07/08, derived from mobile-phone location records and released as a public living-mobility dataset.

Traffic sensors. Two independent sensor sets from Seoul’s TOPIS system: $S_v=121$ point volume sensors (vehicles/hour) and $S_s=396$ link-level speed sensors (km/h), each snapped to the nearest road-graph edge by minimizing Euclidean distance in a projected coordinate system (mean snap distance 46m and 25m respectively, reflecting that speed sensors already carry native link geometry while volume sensors are point counters).

Forecasting task. Given $P=12$ hours of history for every sensor, forecast the next $Q=12$ hours: $\hat{X}_{t+1:t+Q} = \mathcal{F}_\theta(X_{t-P+1:t})$. We report MSE, RMSE, MAE, and MAPE, with a near-zero-denominator guard on MAPE (≥ 10 vehicles/h for volume, ≥ 1 km/h for speed) chosen per-modality because near-zero speed is a genuine congestion state that should not be treated as a sparsity artifact, whereas near-zero volume typically is.

Rolling-origin evaluation. All results use walk-forward validation: 12 months train / 1 month validation / 1 month test, stride 1 month. The sensors’ native 2023–2025 range gives 23 folds; re-aggregating raw monthly files that an upstream script had silently stopped ingesting extends this to 30 folds over 2023–2026.07. Every headline comparison in this paper (Tables 4–8) uses the full 30-fold range. A few secondary diagnostics were run before the extension and keep their original fold counts (the fold-0 capacity sweep of Section 6, the oracle and mechanism (ii)–(v) pilots and the per-sample diagnostic of Section 7); each states its count, and folds from different ranges are never mixed within one significance test.

The injection question. Let $\mathcal{F}_\theta$ be a forecaster taking only past traffic, and $\mathcal{F}'_{\theta,\phi}$ the same architecture augmented with a routed OD signal $od_{t-P+1:t+Q}$ through a fusion module with parameters ϕ (Section 4.3). With enough capacity, ϕ can always be fit to reduce *training* loss, or to ignore an unhelpful signal. The question is whether $\mathcal{F}'$ generalizes significantly better than $\mathcal{F}$ on held-out test months under a paired, walk-forward evaluation – and whether the answer depends on the capacity of $\mathcal{F}_\theta$, which is why we test the same signal across eleven models of differing size.

TABLE 1
Notation used throughout the paper.

Symbol	Meaning
$\mathcal{G}, \mathcal{V}_r, \mathcal{E}_r$	Road network graph, nodes, directed edges
$w_e, \ell_e, \rho_e, \alpha$	Edge weight, length, hierarchy discount, its exponent
$\mathcal{X}_{od}$	Hourly $N \times N$ OD movement tensor
N	Number of OD administrative units (500)
S, S_v, S_s	Sensor count (general, volume = 121, speed = 396)
P, Q	Input / forecast horizon length (12, 12 hours)
W	Static $500 \times S$ routing-weight matrix
$\text{share}(o \rightarrow d)$	Historical destination-share distribution of origin o
od_t, routed_t	Per-origin outflow forecast / routed sensor signal at hour t
$h_t^{\text{traffic}}, h_t^{\text{od}}$	Traffic-branch / OD-branch hidden state at step t
$\beta_t, g(\cdot)$	Per-node/per-step fusion gate, its (MLP) parameterization
θ, ϕ	Base-model / fusion-module parameters

4 METHODOLOGY

4.1 OD Movement Forecasting

The OD forecaster (*HybridOD*) predicts each administrative unit's hourly outflow one step ahead. Its graph connects the $N=500$ units by their historical (training-period) OD flow, so two distant districts joined by a fast subway line are strongly connected while adjacent but poorly connected neighborhoods are not. A graph GRU propagates information over this graph with edge weights modulated by the current hour's calendar features (hour-of-day, day-of-week, and Korean public-holiday indicators for the day and its neighbors), and its hidden state is fused, through the same per-node gate used downstream (Section 4.3), with a calendar-only branch that captures each unit's typical outflow. This design was selected by an architecture search over 14 rolling-origin folds, in which both a pure graph GRU and a pure calendar baseline underperformed the fused model. *HybridOD* is frozen at a single fold-0 checkpoint, so every downstream experiment receives an identical OD forecast and upstream forecast quality cannot vary across folds.

4.2 Road-Network Routing

For each origin unit o , Dijkstra's algorithm on $\mathcal{G}$ (edge weights as in Section 3) gives a shortest-path tree to every other unit; this is run once per origin (not per OD pair), making the full sweep over all $N=500$ origins tractable (≈ 120 seconds on the full 315k-node graph). Each origin's historical (2023 training-period) destination-share distribution $\text{share}(o \rightarrow d)$ is computed from the ground-truth OD tensor. A static routing-weight matrix $W \in \mathbb{R}^{500 \times S}$ is built once:

$$W[o, s] = \sum_d \text{share}(o \rightarrow d) \cdot \mathbb{1}[\text{path}(o, d) \text{ crosses sensor } s's \text{ link}] \quad (2)$$

Applying W to *any* hour's per-origin outflow forecast via a single matrix multiplication $\text{routed}_t = \text{outflow}_t \cdot W$ yields a routed sensor-level signal for that hour. W is a purely structural object – built once from the road network and

the 2023 training-period destination-share distribution – and is reused unchanged across every rolling-origin fold, since it encodes road-network topology and typical destination preference rather than any particular hour's traffic. This routed signal, not a unit's raw outflow pasted onto its nearest sensor, is what we inject: a raw outflow says nothing about *which* roads the trips use, and Section 6.1 confirms quantitatively that the unrouted signal carries no benefit.

Relation to classical traffic assignment. Equation (2) is, in the terminology of the transportation literature, a deterministic *all-or-nothing* network loading: each origin's demand is assigned entirely to shortest paths, with no congestion feedback and no dispersion in route choice. It therefore satisfies neither of Wardrop's equilibrium principles [35] and does not solve the convex program that makes user equilibrium computable [36], [37]. This is deliberate, and we state the tradeoff plainly because a transportation-side reader will notice it immediately. An equilibrium assignment would require link performance functions, a congestion feedback loop, and an iterative solve for every one of the 31,392 hours in the study range, and – more fundamentally – would make the injected signal a function of the very congestion state the downstream model is trying to predict, reintroducing exactly the circularity that a forward-looking demand proxy is meant to avoid. What we need from W is not an accurate flow prediction but a stable structural map from origins to the sensors their outflow plausibly traverses; the learned gate of Section 4.3 is what decides how much to trust the resulting signal, hour by hour and sensor by sensor.

We also tried a stochastic multi-path variant of W (five Dijkstra passes per origin with log-normal edge-weight noise, crediting sensors fractionally), a crude stand-in for Dial's logit assignment [38]. It performed no better than the single-shortest-path W , suggesting that route-choice dispersion is not the binding limitation at this spatial and temporal granularity.

4.3 Gated Fusion Injection (Primary Mechanism)

The traffic model is a graph-recurrent encoder-decoder over a k -nearest-neighbor, Gaussian-kernel-weighted sensor graph (*WeightedGraphGRUCell*), with the kernel bandwidth set adaptively to the median nearest-neighbor distance among sensors. At every decode step t , the routed OD signal is projected through a small MLP, $h_t^{\text{od}} = \text{MLP}(od_t)$, and fused with the traffic branch's hidden state via a per-node, per-timestep learned gate:

$$\beta_t = \sigma\left(g([h_t^{\text{traffic}}; h_t^{\text{od}}])\right), \quad (3)$$

$$h_t^{\text{final}} = \beta_t h_t^{\text{traffic}} + (1 - \beta_t) h_t^{\text{od}}, \quad (4)$$

with the recurrence carried forward at every step by the *pure* traffic branch h_t^{traffic} (not the fused state h_t^{final}), so that the OD signal can modulate the current step's prediction without permanently altering the model's internal traffic-only state trajectory – following the design validated for the OD forecasting model itself. A single global scalar gate (one β shared across all nodes and timesteps within a batch) was tried first and found strictly worse than this per-node/per-step design, confirming that the model needs to learn context-dependent trust in the auxiliary signal rather than a single fixed blending weight.

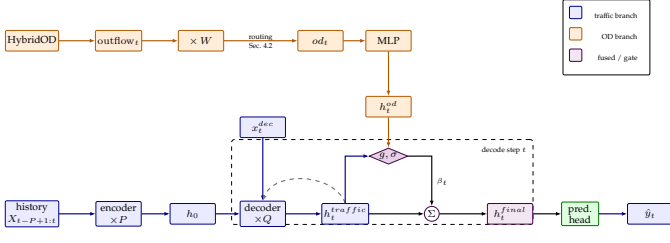


Fig. 1. End-to-end gated-fusion injection architecture (Section 4.3). *Upper left*: the frozen HybridOD forecast is routed onto the road network via the static matrix W (Section 4.2) to produce od_t . *Main flow*: the traffic encoder-decoder consumes recent history; at each decode step the routed signal is projected to h_t^{od} and combined with the traffic branch's $h_t^{traffic}$ through a per-node, per-timestep gate β_t (equation in text). Only the pure $h_t^{traffic}$ (not the fused h_t^{final}) is carried into the next step's recurrence.

5 EXPERIMENTAL SETUP

Datasets. TOPIS traffic volume (121 sensors) and speed (396 sensors), and Seoul capital-region OD movement (500 administrative units), 2023-01 through 2026-07/08, as described in Section 3. Table 2 summarizes basic statistics.

TABLE 2
Dataset summary.

	Volume	Speed
Sensors	121	396
Unit	vehicles/hour	km/h
Sensor type	point counter	link geometry
Mean road-graph snap dist.	46 m	25 m
Date range	2023-01 – 2026-07	
OD administrative units	500	

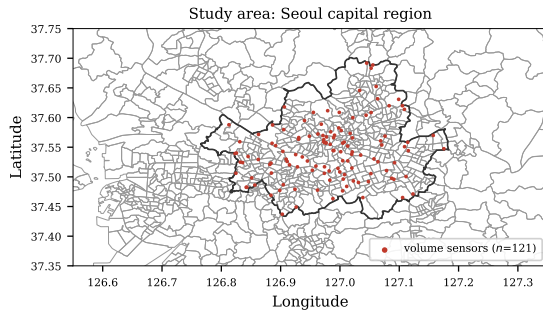


Fig. 2. Study area: Seoul (bold outline) and the surrounding capital region. Thin lines are the 500 OD administrative units (Section 3); red dots are the 121 volume sensors, all within Seoul proper. The 396 speed sensors (link geometries rather than points) are distributed similarly across the same road network and are omitted here for legibility.

Models. Our own lightweight model (Section 4.3; hidden size 48, 37k parameters, 44k with injection) and the ten published architectures of Section 2.1, each reimplemented at the scale of our sensor networks with a shared recipe: $P=Q=12$, Adam, learning rate 2×10^{-3} , up to 40 epochs with early stopping (patience 8) on validation loss, and smooth- L_1 loss (Table 3). Two reimplementations details matter for the results. GMAN's pure self-attention has no

sequential inductive bias and failed to train until we added sinusoidal positional encoding over the lookback axis. A PEMS-BAY fidelity check against published numbers (supplementary Appendix E) shows DCRNN, Graph WaveNet and STAEformer reproducing published accuracy within normal reimplementing variance, and exposed a bug in our first STID, which had replaced its published discrete time-of-day/day-of-week embeddings with continuous features; every STID result reported here uses the corrected architecture. Each architecture is trained in a plain (no-OD) form and with OD injection (Section 7).

TABLE 3

Training hyperparameters. Shared values apply to all models unless overridden.

Hyperparameter	Value
Input / forecast length P, Q	12 / 12 hours
Hidden size	64 (baselines); 48 default (ours)
Optimizer	Adam, lr 2×10^{-3}
Batch size	32 (baselines); 64 (ours)
Max epochs / patience	40 / 8 (15 / 5 for the mechanism-(iv) screen)
Training loss	smooth- L_1
DCRNN diffusion steps	2 (bidirectional)
Graph WaveNet dilation layers	4, kernel size 2
GTS graph-learning temperature	Gumbel-softmax $\tau = 0.5$
k NN graph (ours)	$k = 4$, Gaussian kernel, adaptive bandwidth
Routing hierarchy exponent	0.9
α	
Multipath perturbation (explored)	5 passes, log-normal $\sigma = 0.15$

Evaluation and hardware. We report MSE, RMSE, MAE and MAPE per fold (Section 3) and compare configurations with a paired t -test and a Wilcoxon signed-rank test over the per-fold metric pairs. Adjacent folds share 11 of 12 training months, so folds are not fully independent and p -values should be read as indicative (Section 9). All experiments run on a single NVIDIA H200 GPU. Code, exact fold boundaries, and per-fold results for every reported number are public (footnote, page 1); raw data and checkpoints are omitted for size, but the repository documents each data source.

6 MAIN RESULT: INJECTION ON A LIGHTWEIGHT MODEL

TABLE 4

Traffic-Only vs. Traffic-Hybrid (OD-routed injection), lightweight base model, 30-fold rolling-origin evaluation over the full 2023–2026 extended range (Section 3). Each metric's Δ is (Hybrid – Only) / Only $\times 100\%$, averaged over 30 folds; negative is an improvement. Wins counts folds where Hybrid beats Only on MSE.

Task	Wins	MSE Δ	RMSE Δ	MAE Δ	MAPE Δ
Volume	29/30	–7.17%	–3.69%	–4.02%	–0.06%
Speed	30/30	–8.17%	–4.18%	–4.10%	–3.65%

Both tasks reach paired- t $p < 10^{-5}$ on MSE, RMSE and MAE (speed: $p < 10^{-13}$ on all four metrics), and the Wilcoxon signed-rank test agrees in every case. The volume

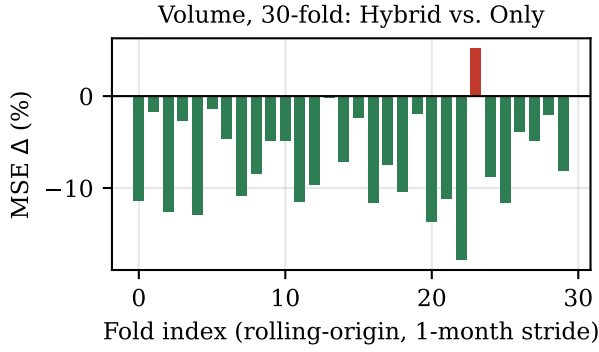


Fig. 3. Per-fold MSE change (Hybrid vs. Only), volume 30-fold set. Green = improvement, red = the single fold that regresses (fold 23, test month 2026-01); every other fold improves.

gain is not driven by a few large wins: 29 of 30 folds improve, and the single regression (fold 23, test month 2026-01) is small (Figure 3). Volume MAPE barely moves (-0.06%) because MAPE weights low-volume off-peak hours as heavily as rush hours, whereas MSE is dominated by the high-volume errors where the gain concentrates.

Absolute performance vs. published architectures. Our base model is deliberately lightweight and does not match DCRNN or Graph WaveNet in absolute terms: at fold 0, Traffic-Only reaches 178,251 MSE and Traffic-Hybrid 157,937, against 128,014 for plain DCRNN. The claim is not that injection makes a small model state-of-the-art, but that a cheap auxiliary signal closes 40.4% of the gap to a considerably stronger architecture with no change in capacity – a partial substitute for capacity, most useful where capacity is constrained.

Capacity sweep. To see how the injection effect interacts with capacity, we retrained Traffic-Only/Hybrid at hidden sizes 64 and 96 on fold 0. Both improve with capacity (Only: 178,251 $\rightarrow$ 165,573 $\rightarrow$ 129,782; Hybrid: 157,937 $\rightarrow$ 147,060 $\rightarrow$ 127,629), and at hidden 96 Traffic-Hybrid matches plain DCRNN (-0.3%). But the injection effect itself does not stay constant: it is -11.4% at hidden 48, -11.2% at 64, and collapses to -1.7% at 96 (Figure 4). As the model gains enough capacity to match a far more sophisticated architecture on its own, the benefit of the routed signal shrinks toward zero – the first sign of the capacity gap that Section 7 establishes across architectures. This is a single-fold, single-seed trend; Section 7.3 repeats the test at full 30-fold resolution on a published architecture.

Why the effect is larger for speed. The speed gain (-8.2%) consistently exceeds the volume gain (-7.2%) although both share the same OD signal, routing and fusion. Two explanations are plausible and not mutually exclusive. The speed network is three times denser and its sensors carry native link geometry (Section 3), so its routing matrix is better localized. And volume saturates at a road’s physical capacity while speed degrades continuously under an unanticipated demand surge, leaving forward-looking demand information more headroom on speed. A sensor-density-matched comparison that could separate the two is not possible with Seoul’s deployed network.

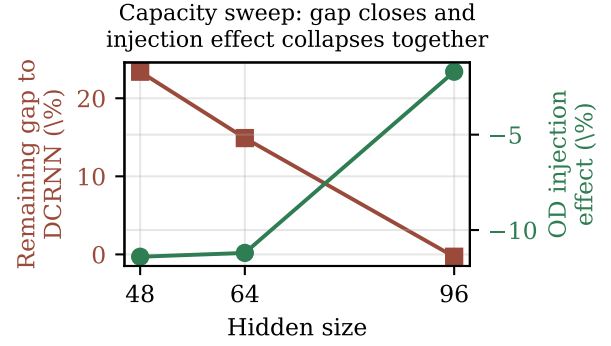


Fig. 4. Capacity sweep (hidden 48/64/96): the gap to DCRNN (red, left axis) closes and the OD-injection effect (green, right axis) collapses toward zero in lockstep as capacity increases.

6.1 Robustness Checks: Routing Necessity and Calendar Confounding

We test two threats to the headline result directly, holding architecture, fusion mechanism and training recipe fixed and varying only the injected signal, on the same 30-fold volume sweep (Table 5).

TABLE 5

Signal controls on the headline model (lightweight model, volume, 30 folds). Each row injects a different signal through the identical fusion mechanism; Δ is MSE change versus Traffic-Only. Recovery is each row’s Δ as a fraction of the real routed signal’s.

Injected signal	Wins	Mean Δ	p	Rec.
Real routed OD (Hybrid)	29/30	-7.17%	8.0e-08	–
Unrouted OD (no $\times W$)	20/30	-0.39%	0.61	5%
Calendar climatology	27/30	-6.22%	2.9e-06	87%
Residual (routed – clim.)	24/30	-2.47%	0.019	34%
Climatology + residual	25/30	-3.88%	1.9e-04	54%

Is routing necessary? Removing the road-network projection $\times W$ leaves an OD signal indistinguishable from no injection (-0.39% , $p = 0.61$). The benefit is not produced by “any OD-shaped channel”: the routing step does real work.

Calendar confounding. The routed signal has a strong daily and weekly rhythm, so part of its benefit could come from the model learning a calendar prior rather than forward-looking OD content. For each fold we build a (day-of-week, hour-of-day) climatology of the routed signal from that fold’s *training window only* – a lookup table with no OD model and no forecast, which correlates with the real signal at $r = 0.997$. Injected alone, it recovers 87% of the gain: most of the average improvement comes from a calendar prior learned through the injection pathway.

Isolating the genuine OD content. Subtracting each fold’s climatology from the real signal, $\text{residual}_t = \text{real}_t - \text{climatology}[\text{bucket}(t)]$, leaves a signal with no calendar information by construction. Injected alone, it still recovers a significant 34% of the gain: day-specific OD content contributes, but as the secondary share. Feeding climatology and residual as two separately normalized channels lands in between (54%), so the natural quantity – real routed movement, climatology plus residual – is also the best one to inject.

All four controls use the 30-fold volume sweep; they have not yet been run on speed.

7 DOES INJECTION GENERALIZE TO STRONGER ARCHITECTURES?

A positive result on one lightweight architecture may not generalize. We therefore give the identical routed OD signal (Section 4.2) to the ten published architectures of Section 2.1, which span 31k–774k parameters – STID is smaller than our own 37k-parameter model, the other nine are larger – and cover fixed-diffusion, learned-adjacency, attention-based and graph-free designs. The main mechanism is the same gate our lightweight model uses; four alternatives test whether a failure could be blamed on the particular fusion design:

- (i) *Node-feature gated fusion* (main mechanism): the per-step gate of Section 4.3, applied to each architecture’s decoder representation, with an explicit future-calendar term.¹
- (ii) *Graph-adjacency modulation*: a per-node OD-activity gate rescales the architecture’s own fixed (DCRNN) or adaptive (Graph WaveNet, GTS) adjacency, injecting OD into *where information flows* rather than as a node feature.
- (iii) *Autoregressive decoding* (DCRNN): a true Q -step autoregressive decoder with the gate at every step, mirroring our own model exactly.
- (iv) *Boost/eliminate dual gate*: the knowledge-adaptation gate of [22], which up-weights hidden dimensions the auxiliary signal corroborates and suppresses those it contradicts – strictly more expressive than (i).
- (v) *Cross-attention* (STAEformer): each decode step attends over the OD signal at all Q future steps.

7.1 Uniform Injection Across Ten Architectures

TABLE 6

OD injection (mechanism (i), node-feature gated fusion) across ten published or reimplemented architectures, ordered by parameter count. All rows are the extended 30-fold rolling-origin sweep. [†] = significant in the *favorable* (error-reducing) direction; all other bolded entries are significant and unfavorable. The two smallest architectures are the only ones that benefit; the effect turns null near 10^5 parameters and significantly harmful above it.

Model	Params	Wins	Volume Mean Δ (p)	Wins	Speed Mean Δ (p)
STID	31,028	30/30	−9.02%[†] (1.7e-10)	11/30	+0.24% (0.57)
GMAN	54,748	22/30	−7.53%[†] (0.002)	14/30	+0.19% (0.789)
STGCN	62,236	20/30	−2.09% (0.831)	16/30	+0.09% (0.925)
PDFormer	94,988	23/30	−2.70% (0.274)	2/30	+3.25% (1e-8)
STAEformer	159,996	14/30	+0.74% (0.484)	1/30	+4.71% (5e-10)
MTGNN	335,452	3/30	+11.07% (5e-5)	0/30	+4.24% (7e-14)
DCRNN	381,916	14/30	+0.95% (0.254)	11/30	+1.76% (0.011)
GTS	394,853	16/30	−0.19% (0.877)	3/30	+3.86% (5e-9)
GWNET	672,656	1/30	+25.45% (4e-9)	5/30	+3.77% (1e-5)
AGCRN	773,830	3/30	+11.92% (5e-6)	11/30	+0.91% (0.027)

The effect is ordered by capacity. Table 6 and Figure 5a–b show the main mechanism on all ten architectures, ordered

1. An early version without the calendar term was statistically indistinguishable from injecting temporally shuffled OD.

by parameter count. Only the two smallest benefit significantly, both on volume only: STID (−9.02%, 30/30 folds, $p = 1.7 \times 10^{-10}$) and GMAN (−7.53%, 22/30, $p = 0.002$). STGCN and PDFormer, near 10^5 parameters, are null on volume. Every architecture above 10^5 parameters is significantly *harmful* on at least one task – ten of the twenty (architecture, task) combinations in total – and the harm is largest for the largest models (Graph WaveNet +25.45%, AGCRN +11.92%, MTGNN +11.07% on volume). Across all twenty points the effect correlates with $\log_{10}$ parameters at Spearman $\rho = 0.67$ ($p = 1.2 \times 10^{-3}$), and at $\rho = 0.89$ ($p = 5.4 \times 10^{-4}$) on volume alone. Speed is less cleanly ordered ($\rho = 0.45$): no small model benefits there, so capacity is a strong but not complete predictor, and we flag the volume/speed asymmetry as open. GMAN’s favorable volume result is not a lucky seed – seeds 0, 1 and 2 give −7.53%, −13.78% and −9.96%, all significant, and we report the most conservative.

It is not the fusion design. The alternative mechanisms show the same null-to-negative pattern. Graph-adjacency modulation on Graph WaveNet is null (+1.17%, $p = 0.52$, 23 native-range folds). DCRNN’s autoregressive decoder is worse with OD than without it at fold 0 (140,689 vs. 134,693 MSE). The more expressive boost/eliminate gate, in a 6-fold warm-started screen, averages +2.32% (median −0.33%, two folds +6.7% and +10.1%). Cross-attention on STAEformer is +29.1% worse at fold 0, with visibly destabilized training, and was stopped there.

It is not forecast error. Injecting the *true*, already-observed OD instead of a forecast into DCRNN still gives no improvement (+1.31%, 4/9 native-range folds). Even a perfect OD forecast would not have helped, so a better forecaster is not the missing piece.

Why single-fold results mislead. At fold 0 alone, the main mechanism on STAEformer looked like a breakthrough (−3.67%), and a single-split study would have reported it as such. Over 30 folds it is noise (+0.74%, 14/30, $p = 0.48$) – a concrete example of the risk the rolling-origin protocol guards against.

We call this pattern the *capacity gap*: once an architecture can infer from traffic history most of what the routed signal would tell it, a fixed fusion mechanism has little to gain and enough spare capacity to fit the signal’s imperfections. One objection remains: uniform injection measures the signal through one particular fusion mechanism. The signal might still carry value for the larger architectures that their fusion mechanism simply fails to extract. The next subsection tests this without relying on any fusion mechanism at all.

7.2 Does the Signal Add Value Beyond Ensembling?

If the injected checkpoint carries information the plain one lacks, that information should be recoverable *after* training by combining the two – for example, by averaging their predictions, or by choosing per sensor whichever is better on the validation month (selective injection). But any combination of two checkpoints also benefits from ordinary ensembling, whether or not either carries new information. A fair test must therefore compare the injected checkpoint against the right control: a second *plain* checkpoint differing only in random seed. For every model, task and fold we compute

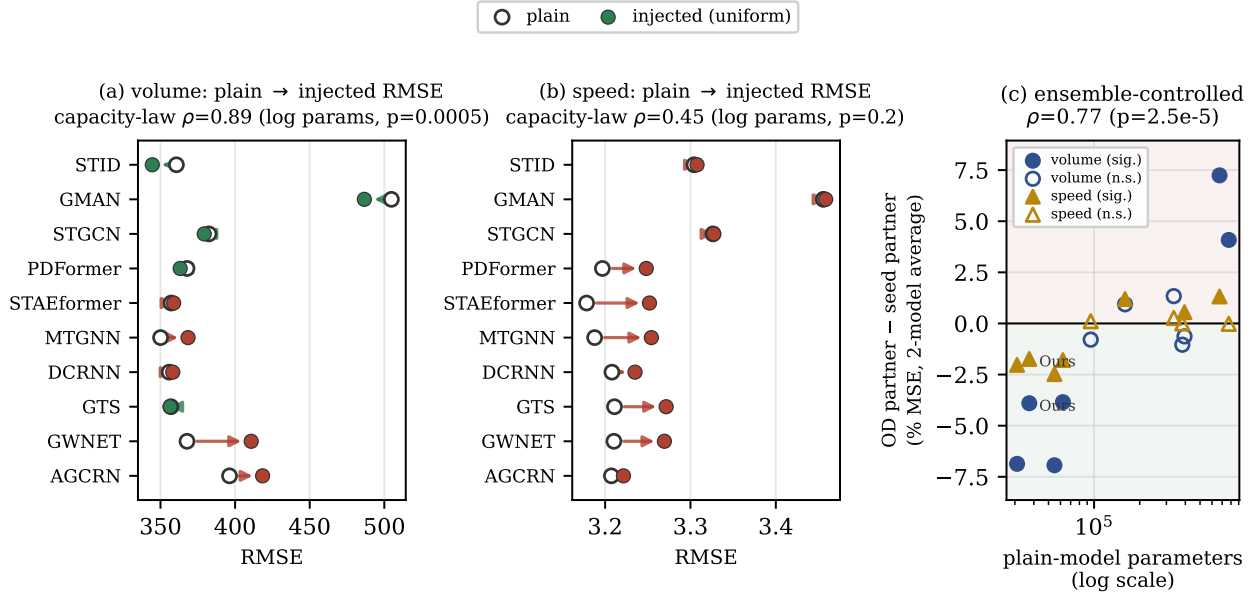


Fig. 5. Injection effect versus model capacity. (a) Volume and (b) speed: absolute RMSE of each of the ten published architectures, plain (open circle) and with uniform OD injection (filled circle; green improves, red worsens), ordered by parameter count (smallest at top); the effect correlates with $\log_{10}$ parameters at Spearman $\rho = 0.89$ on volume ($\rho = 0.67$ pooled, $p = 1.2 \times 10^{-3}$). (c) Ensemble-controlled test (Table 7): how much better the injected model is as an ensemble partner than a second plain seed, for all eleven models; filled markers are significant after Benjamini-Hochberg correction. Below zero, the OD signal adds value beyond ensembling; above zero, it costs accuracy. The crossover lies near 10^5 parameters ($\rho = 0.77$, $p = 2.5 \times 10^{-5}$).

Algorithm 1 Ensemble-controlled test of an auxiliary signal

Input: architecture $\mathcal{F}$; auxiliary signal od ; rolling-origin folds $k = 1, \dots, K$, each with train, validation and test months.
Output: per-fold OD partner gain g_k and its significance.

- 1) For each fold k , train three checkpoints on its train month range, with early stopping on its validation month: plain P_k (seed 0), plain P'_k (seed 1), and injected I_k (seed 0, with od).
- 2) On the test month, form the two-model averages $\hat{y}_k^{PI} = \frac{1}{2}(\hat{y}_k^P + \hat{y}_k^I)$ and $\hat{y}_k^{PP'} = \frac{1}{2}(\hat{y}_k^P + \hat{y}_k^{P'})$.
- 3) Compute the relative MSE changes against P_k and their difference: $g_k = \Delta\text{MSE}(\hat{y}_k^{PI}) - \Delta\text{MSE}(\hat{y}_k^{PP'})$.
- 4) Test $\mathbb{E}[g_k] = 0$ with a paired t -test over folds; correct across all (architecture, task) pairs with Benjamini-Hochberg.

$g < 0$: the signal adds value beyond ensembling. $g \approx 0$: the injected model is only another ensemble member. $g > 0$: injecting the signal costs accuracy even after averaging.

the test MSE of the plain model averaged with the injected model, $\text{Avg}(P, I)$, and of the plain model averaged with its second seed, $\text{Avg}(P, P')$. Their difference – the *OD partner gain* – isolates what the OD signal contributes beyond ensembling. It needs no fusion mechanism, no selection rule, and no new training beyond the second seed (Algorithm 1, Table 7).

The capacity gap survives ensemble control, and becomes sharper. For every model up to 62k parameters – STID, our own model, GMAN and STGCN – the injected model is a significantly better ensemble partner than a second seed, on *both* tasks (−1.7% to −6.9%, all $q \leq 0.01$). This includes the speed task, where uniform injection had shown no benefit for STID, GMAN or STGCN: the signal was there, but the injected model alone could not turn it into a net gain. For every model of 95k parameters or

more, the injected partner is at best equivalent to a second seed (nine combinations, none significant) and in five combinations significantly *worse* (Graph WaveNet on both tasks, AGCRN/volume, STAEformer/speed, GTS/speed). No large model gains anything from the OD-injected partner that a second seed would not give it. The OD partner gain correlates with $\log_{10}$ parameters at $\rho = 0.77$ ($p = 2.5 \times 10^{-5}$) across all 22 points, with the speed task now significant on its own as well ($\rho = 0.67$, $p = 0.023$, vs. $\rho = 0.45$ under uniform injection; Figure 5c).

What this means for combining checkpoints. Two-model averaging is a strong baseline in its own right: $\text{Avg}(P, P')$ alone reduces MSE by 2–12% with no OD signal at all. Selective injection, which picks one checkpoint per sensor rather than averaging, never exploits this variance reduction and is beaten by $\text{Avg}(P, I)$ in 21 of 22 combinations; its apparent recovery of the large architectures is largely the ordinary benefit of combining two checkpoints (supplementary Appendix B reports it in full, with a seed-only selection control). The practical implication is therefore simple. For a lightweight model, the injected checkpoint is worth training – alone (−7.2%/−8.2% for ours) or, better, averaged with the plain one (−11.6%/−8.5%). For a model of 10^5 parameters or more, the same compute is better spent on a second plain seed.

7.3 Within-Architecture Capacity Sweep on a Published Model

The between-architecture correlation is suggestive but confounded: architectures differ in far more than parameter count. The cleanest test varies capacity alone inside one published architecture. We use STID, which is the smallest

TABLE 7

Does the OD-injected model add value beyond ensembling? MSE change versus the plain model (30 folds; V/S = volume/speed). *Uniform*: the injected model alone (Table 6). *Selective*: per-sensor validation-gated choice between plain and injected. *Avg(P, I)*: average of the plain and injected predictions. *Avg(P, P')*: average of the plain model and a second plain model differing only in seed – the gain any two-model average obtains without OD. *OD partner gain* = *Avg(P, I)* – *Avg(P, P')*: negative means the injected model is a better ensemble partner than another seed; *q* is a paired *t*-test with Benjamini–Hochberg correction across all 22 rows. Bold = significant. The rule separates models up to 62k parameters from the larger ones.

Model	Params	Task	Uniform	Selective	Avg(P, I)	Avg(P, P')	OD partner gain	<i>q</i>
STID	31k	V	−9.02%	−10.00%	−13.19%	−6.33%	−6.87%	<0.001
		S	+0.24%	−3.39%	−4.24%	−2.22%	−2.02%	<0.001
Ours	37k	V	−7.17%	−9.41%	−11.62%	−7.72%	−3.90%	<0.001
		S	−8.17%	−9.23%	−8.47%	−6.73%	−1.73%	<0.001
GMAN	55k	V	−7.53%	−14.14%	−18.10%	−11.17%	−6.94%	<0.001
		S	+0.19%	−5.54%	−6.82%	−4.34%	−2.48%	<0.001
STGCN	62k	V	−2.09%	−7.84%	−13.90%	−10.05%	−3.85%	0.010
		S	+0.09%	−3.73%	−5.45%	−3.66%	−1.79%	<0.001
PDFormer	95k	V	−2.70%	−6.73%	−12.11%	−11.32%	−0.79%	0.593
		S	+3.25%	−1.65%	−3.14%	−3.25%	+0.11%	0.626
STAEformer	160k	V	+0.74%	−4.49%	−10.69%	−11.62%	+0.94%	0.236
		S	+4.71%	−1.23%	−2.93%	−4.12%	+1.19%	<0.001
MTGNN	335k	V	+11.07%	−1.37%	−6.98%	−8.31%	+1.34%	0.213
		S	+4.24%	−1.71%	−3.13%	−3.40%	+0.27%	0.465
DCRNN	382k	V	+0.95%	−4.06%	−9.92%	−8.88%	−1.04%	0.465
		S	+1.76%	−1.51%	−3.50%	−3.50%	+0.00%	0.932
GTS	395k	V	−0.19%	−5.10%	−10.86%	−10.23%	−0.63%	0.626
		S	+3.86%	−1.44%	−3.17%	−3.72%	+0.55%	0.010
GWNEN	673k	V	+25.45%	−0.46%	−2.83%	−10.08%	+7.24%	<0.001
		S	+3.77%	−1.87%	−4.29%	−5.61%	+1.32%	<0.001
AGCRN	774k	V	+11.92%	−1.86%	−6.97%	−11.06%	+4.08%	<0.001
		S	+0.91%	−1.45%	−3.88%	−3.86%	−0.02%	0.909

architecture in the study, has the largest favorable effect, and – unlike GMAN – is already a healthy model at its default size (plain volume MSE 132,418 vs. GMAN’s 260,733), so a collapse of the effect at higher capacity cannot be explained by a poorly trained small baseline finally converging. We retrain STID at hidden sizes 64, 128 and 256 (31k to 419k parameters), 30 folds each, with everything else fixed.

TABLE 8

STID capacity sweep, volume, 30 folds per row, corrected architecture throughout. Injection’s benefit decays monotonically with capacity, loses significance by hidden 128, and reverses to significantly harmful by hidden 256, while the plain model’s own accuracy saturates.

Hidden	Params	Plain MSE	Injection Δ	<i>p</i>
64 (seed 0)	31,028	132,418	−9.02%	1.7e-10
64 (seed 1)	31,028	132,102	−6.76%	1.4e-6
128	111,196	124,190	−2.30%	0.347
256	418,988	122,712	+3.37%	0.014

A $13\times$ increase in capacity carries the injection effect not just toward zero but through it (Table 8, Figure 5c): from $−9.02\%$ ($p = 1.7 \times 10^{-10}$) at hidden 64, to null at 128, to significantly *harmful* at 256 ($+3.37\%$, $p = 0.014$) – the same sign reversal Table 6 shows between architectures, now produced by scaling a single one. Two seeds at hidden 64 both reach significance ($−9.02\%$, $−6.76\%$), so the curve’s starting point is not a seed artifact. Meanwhile the plain model’s own accuracy improves and saturates ($132,418 \rightarrow 124,190 \rightarrow 122,712$): the signal stops helping, and starts hurting, exactly as the model stops needing it and gains the spare capacity to fit its imperfections. This is what a redundancy account predicts and an independent-information account does not. With the fold-0 sweep on our

own model (Section 6), the pattern now holds in two model families.

7.4 Beyond Seoul: Replication on Public Benchmarks

Every result so far comes from one city and one signal. To test whether the capacity law is a property of signal injection rather than of our dataset, we repeat the between-architecture test on three standard public benchmarks – PEMS-BAY (325 sensors), METR-LA (207) and PeMSD7 (228; 44 weekdays) – aggregated to hourly resolution with Seoul’s 12-hour history and horizon. These datasets have no OD data, so we inject a pure calendar climatology of each sensor’s own traffic through the same ten architectures: a harder test, since the injected signal carries no forecast content at all (protocol details in supplementary Appendix C).

TABLE 9

Capacity-law replication across four independent datasets: Spearman correlation between $\log_{10}(\text{parameters})$ and injection effect across the same ten architectures, hourly resolution, horizon matched to Seoul (12h $\rightarrow$ 12h).

Dataset	Signal	ρ	<i>p</i>
Seoul (this paper)	routed OD forecast	+0.891	5.4e-4
PEMS-BAY	calendar climatology	+0.830	0.003
PeMSD7	calendar climatology	+0.503	0.138
METR-LA	calendar climatology	+0.370	0.293

All four correlations are positive: under a null of no effect, four independent datasets landing on the same side has probability $1/16$ by sign alone. The law reaches significance on Seoul and PEMS-BAY, the two datasets with the

longest and cleanest records, but not on PeMSD7 (44 days) or METR-LA (the noisiest, 8% missing at native resolution). We read this as replication in direction everywhere and in significance where data quality allows: the capacity gap is not specific to Seoul or to OD, and its statistical strength, rather than its existence, depends on the host dataset.

7.5 Diagnostic: When Does the Signal Help?

Aggregating ~ 280 million per-sample predictions across DCRNN’s 23-fold (native-range) volume sweep, we bin samples into quintiles by how far the injected OD value deviates from its own (day-of-week, hour-of-day) typical baseline – computed from the training window only – normalized per sensor and pooled across all 121 sensors and 23 folds, and measure mean error reduction (Only – Hybrid) per quintile. A decile-resolution repeat reproduces the same monotonic pattern within sampling noise.

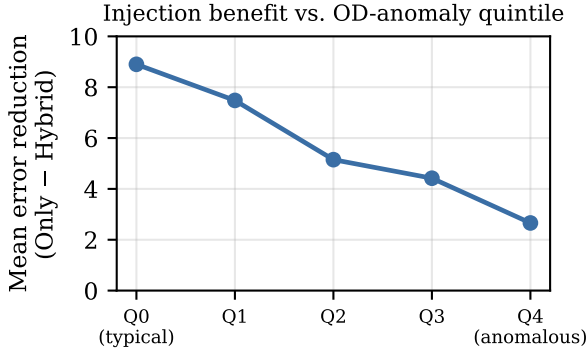


Fig. 6. Mean error reduction by OD-anomaly quintile: largest when the injected signal is closest to typical (Q0), smallest when it is most anomalous (Q4) – the opposite of the intuition that motivated this diagnostic.

Improvement is monotonically *largest* on the most typical samples and smallest on the most anomalous ones (Figure 6) – the opposite of the intuition that OD’s value should lie in capturing surprises that past-traffic autocorrelation alone cannot. We read this as follows: the injected signal is itself a *forecast*, most accurate exactly when the pattern is typical. On typical hours it correctly confirms what a sufficiently strong model already infers – redundant but low-cost. On anomalous hours – the one regime where new information would matter most – the OD forecast is itself least reliable, so it cannot compensate for what autocorrelation misses.

Figure 7 makes this concrete for our own model at fold 0. For a single sensor (panel a), the hour-by-hour comparison is noisy – the injected model wins 90 of 168 test hours – but it roughly halves the error at the week’s largest spike, a Monday 07:00 commute surge. Averaged over all 121 sensors (panel b), the pattern is clean: the error reduction peaks at the morning commute (07:00–08:00, +43 and +40 vehicles/h) with a secondary evening peak (17:00, +29), and is smallest overnight. Commute peaks are the hours when OD movement most directly and predictably produces road traffic, and, across many weekdays, a *typical* rather than anomalous pattern – the regime Figure 6 identifies as the one where the injected signal helps.

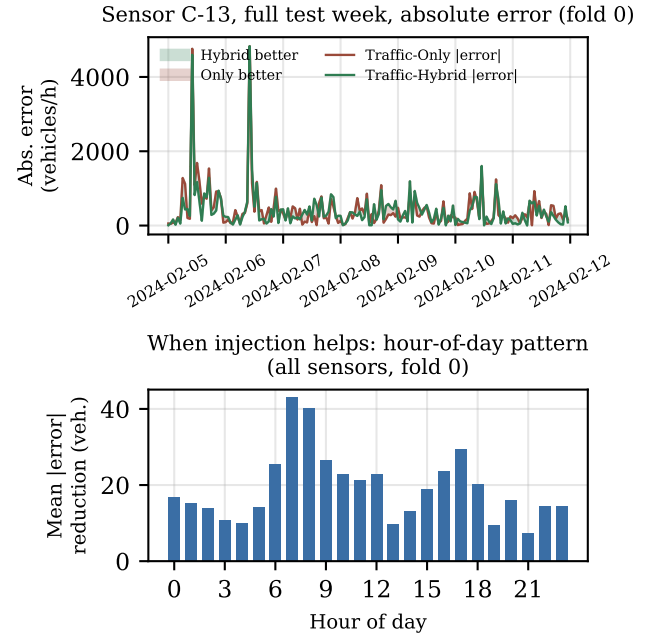


Fig. 7. Qualitative case study, fold 0: (a) one sensor’s absolute error, full test week (not cropped) – green/red shading shows which variant wins each hour; noisy hour-to-hour (Hybrid wins 90/168 hours) but Hybrid clearly wins the week’s largest spikes; (b) the same error-reduction quantity aggregated over all 121 sensors, where the commute-peak pattern is unambiguous.

A secondary check correlating each sensor’s improvement with its own spatial autocorrelation (hypothesizing that injection should help most at spatially-hard-to-reason-about sensors) found only a weak positive association ($r = 0.161$; weak/mid/strong terciles improve +4.51/ +6.78/ +5.98), more plausibly explained by well-connected sensors having denser OD-routing coverage in W than by a capacity effect at the sensor level.

8 DISCUSSION

Why a capacity gap? The three tests point to one mechanism. The routed OD signal is itself a forecast, built from the same daily and weekly regularities that drive traffic, so most of what it says is predictable from traffic history alone (Section 6.1: 87% of its benefit is reproduced by a calendar lookup). A small model cannot represent those regularities by itself and gains from being told them; a large model already represents them and gains nothing, while its spare capacity lets it fit the signal’s imperfections – the forecast errors that concentrate in exactly the anomalous hours where new information would matter most (Section 7.5). This explains both the sign reversal under uniform injection and why even a perfect (oracle) OD signal does not help DCRNN: the value of an auxiliary signal is bounded by what the model does not already know, not by the signal’s accuracy.

Implications for auxiliary-knowledge injection. Many spatiotemporal methods inject external knowledge – points of interest, weather, events, regional embeddings, or pre-trained representations – and evaluate the gain on one

or two architectures. Our results suggest that such gains should be reported as a function of base-model capacity, and that ensemble effects must be controlled before attributing a gain to the injected knowledge: in our data, averaging two checkpoints without any signal already reduces MSE by 2–12%, a gain easily mistaken for the value of injected knowledge. Algorithm 1 is a simple, architecture-agnostic way to do both.

Cost. The routing matrix W is built once (≈ 2 minutes for 500 origins on the full 315k-node graph) and applied as one matrix product per hour. Injection adds 19% parameters to our lightweight model (37k $\rightarrow$ 44k). The ensemble-controlled test costs one additional training run per fold; for a lightweight model, averaging the plain and injected checkpoints doubles inference cost for a further 4–5 points of MSE reduction over injection alone on volume.

9 LIMITATIONS AND THREATS TO VALIDITY

Most of the lightweight-model gain is calendar-shaped. A training-window climatology of the routed signal recovers 87% of the uniform-injection gain, and the calendar-free residual alone recovers 34% (Section 6.1). Day-specific OD content contributes, but as the secondary share, and these controls have so far been run on volume only.

The ensemble-controlled test uses one second seed. The OD partner gain compares the injected model with a single additional plain seed per fold. Seed-to-seed variation differs across architectures (1.4%–9.7% MSE), so the comparison is noisier for high-variance models such as GMAN; 30 paired folds and Benjamini–Hochberg correction guard against over-reading any single combination, but a multi-seed version would tighten the estimates.

Fold non-independence. Adjacent folds share 11 of 12 training months, so the paired-test p -values are likely somewhat optimistic. We therefore read them together with the per-fold consistency of each result (e.g. win counts, Figure 3), not in isolation.

Static routing and a frozen forecaster. The routing matrix W is built once from 2023 destination shares, and the OD forecaster is frozen at a fold-0 checkpoint; neither is updated for later folds. Road topology changes little over this period, but shifts in destination preference and forecaster staleness could both understate the injection effect in later folds, and may contribute to the diagnostic of Section 7.5.

Scope. All OD experiments use one city, one OD data source and one family of fusion mechanisms (gated or adjacency-level fusion of a routed scalar). The capacity law replicates in direction on three public benchmarks, but with a calendar signal rather than OD, and reaches significance on only one of them (Section 7.4). Mechanism pilots (iii)–(v) and the oracle were run on fewer folds, one of them (mechanism iv) with warm-started rather than independent per-fold training, and are reported as supporting rather than confirmatory evidence.

10 CONCLUSION

Forecast population movement, routed onto the road network, does carry information that traffic models can use – but only models small enough to lack it on their own.

Injected through a learned gate, routed OD improves a lightweight graph-recurrent model by 7.2% (volume) and 8.2% (speed) in MSE, closing 40% of its gap to DCRNN; road-network routing is essential, and most of the gain is calendar-shaped. Across ten published architectures, the same mechanism helps only the two smallest and harms every architecture above 10^5 parameters on at least one task, in an order that tracks model size and that scaling a single architecture reproduces. An oracle experiment and alternative fusion designs rule out forecast error and fusion expressiveness as the explanation, and a per-sample diagnostic shows the signal is most reliable exactly where a strong model needs it least.

A fusion-free, ensemble-controlled test removes the last alternative explanation: as an ensemble partner, the OD-injected model beats a second random seed for every model up to 62k parameters, on both tasks, and never for any model of 95k parameters or more. The same test shows that per-sensor selection between checkpoints, a seemingly effective remedy for the larger models, owes most of its gain to ordinary ensembling. For practitioners the recommendation is therefore simple: inject auxiliary mobility signals into compact models, preferably averaging the injected and plain checkpoints; for large models, spend the same compute on a second seed. For researchers, two directions follow. The anomalous hours where OD forecasts are least reliable (Figure 6) are where genuinely new information would matter most, calling for event- or anomaly-conditioned injection rather than a plain point forecast. And the capacity gap should be tested in other upstream-cause/downstream-effect settings – weather in energy demand, flight schedules in airport ground traffic – to establish whether it is specific to OD-to-traffic injection or a general property of auxiliary-signal injection, with the ensemble-controlled protocol as the test.

REFERENCES

- [1] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2018.
- [2] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in *International Conference on Learning Representations (ICLR)*, 2018.
- [3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph wavenet for deep spatial-temporal graph modeling,” in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2019.
- [4] S. Guo, Y. Lin, N. Feng, C. Song, and H. Wan, “Attention based spatial-temporal graph convolutional networks for traffic flow forecasting,” in *AAAI Conference on Artificial Intelligence*, 2019.
- [5] C. Zheng, X. Fan, C. Wang, and J. Qi, “Gman: A graph multi-attention network for traffic prediction,” in *AAAI Conference on Artificial Intelligence*, 2020.
- [6] H. Liu, Z. Dong, R. Jiang, J. Deng, Q. Chen, and X. Song, “Staformer: Spatio-temporal adaptive embedding makes vanilla transformer sota for traffic forecasting,” in *ACM International Conference on Information and Knowledge Management (CIKM)*, 2023.
- [7] L. Bai, L. Yao, C. Li, X. Wang, and C. Wang, “Adaptive graph convolutional recurrent network for traffic forecasting,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [8] C. Shang, J. Chen, and J. Bi, “Discrete graph structure learning for forecasting multiple time series,” in *International Conference on Learning Representations (ICLR)*, 2021.
- [9] Z. Wu, S. Pan, G. Long, J. Jiang, X. Chang, and C. Zhang, “Connecting the dots: Multivariate time series forecasting with graph neural networks,” in *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2020.

- [10] S. Han, J. An, and D. Lee, "Spatio-temporal road traffic prediction using real-time regional knowledge," *arXiv preprint arXiv:2408.12882*, 2024.
- [11] Z. Shao, Z. Zhang, F. Wang, W. Wei, and Y. Xu, "Spatial-temporal identity: A simple yet effective baseline for multivariate time series forecasting," in *ACM International Conference on Information and Knowledge Management (CIKM)*, 2022, pp. 4454–4458.
- [12] J. Jiang, C. Han, W. X. Zhao, and J. Wang, "Pdformer: Propagating delay-aware dynamic long-range transformer for traffic flow prediction," in *AAAI Conference on Artificial Intelligence*, 2023.
- [13] G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*. Holden-Day, 1970.
- [14] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.
- [15] S. Guo, Y. Lin, H. Wan, X. Li, and G. Cong, "Learning dynamics and heterogeneity of spatial-temporal graph data for traffic forecasting," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 11, pp. 5415–5428, 2022.
- [16] C. Song, Y. Lin, S. Guo, and H. Wan, "Spatial-temporal synchronous graph convolutional networks: A new framework for spatial-temporal network data forecasting," in *AAAI Conference on Artificial Intelligence*, 2020.
- [17] L. Zhao, Y. Song, C. Zhang, Y. Liu, P. Wang, T. Lin, M. Deng, and H. Li, "T-gcn: A temporal graph convolutional network for traffic prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 9, pp. 3848–3858, 2020.
- [18] J. Zhang, Y. Zheng, and D. Qi, "Deep spatio-temporal residual networks for citywide crowd flows prediction," in *AAAI Conference on Artificial Intelligence*, 2017.
- [19] Z. Shao, Z. Zhang, F. Wang, W. Wei, and Y. Xu, "Decoupled dynamic spatial-temporal graph neural network for traffic forecasting," *Proceedings of the VLDB Endowment*, vol. 15, no. 11, 2022.
- [20] J. Han, W. Zhang, H. Liu, T. Tao, N. Tan, and H. Xiong, "Bigst: Linear complexity spatio-temporal graph neural network for traffic forecasting on large-scale road networks," *Proceedings of the VLDB Endowment*, vol. 17, no. 5, pp. 1081–1090, 2024.
- [21] S. Han, J. An, and D. Lee, "Urban vibrancy embedding and application on traffic prediction," *arXiv preprint arXiv:2602.21232*, 2026.
- [22] C. Li, L. Bai, W. Liu, L. Yao, and S. T. Waller, "Knowledge adaption for demand prediction based on multi-task memory neural network," in *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020, pp. 715–724.
- [23] L. Wang, X. Geng, X. Ma, F. Liu, and Q. Yang, "Cross-city transfer learning for deep spatio-temporal prediction," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2019.
- [24] Y. Jin, K. Chen, and Q. Yang, "Selective cross-city transfer learning for traffic prediction via source city region re-weighting," in *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2022.
- [25] Y. Wei, Y. Zheng, and Q. Yang, "Transfer knowledge between cities," in *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016.
- [26] Y. Yuan, J. Ding, J. Feng, D. Jin, and Y. Li, "A universal pre-training and prompting framework for general urban spatio-temporal prediction," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 5, pp. 2212–2225, 2025.
- [27] Z. Li, L. Xia, Y. Xu, and C. Huang, "Gpt-st: Generative pre-training of spatio-temporal graph neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [28] Z. Shao, Z. Zhang, F. Wang, and Y. Xu, "Pre-training enhanced spatial-temporal graph neural network for multivariate time series forecasting," in *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2022.
- [29] Z. Li, L. Xia, J. Tang, Y. Xu, L. Shi, L. Xia, D. Yin, and C. Huang, "Urbangpt: Spatio-temporal large language models," in *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024.
- [30] C. Liu, S. Yang, Q. Xu, Z. Li, C. Long, Z. Li, and R. Zhao, "Spatial-temporal large language model for traffic prediction," *arXiv preprint arXiv:2401.10134*, 2024.
- [31] Z. Li, L. Xia, L. Shi, Y. Xu, D. Yin, and C. Huang, "Opencity: Open spatio-temporal foundation models for traffic prediction," *arXiv preprint arXiv:2408.10269*, 2024.
- [32] Z. Shao, F. Wang, Y. Xu, W. Wei, C. Yu, Z. Zhang, D. Yao, T. Sun, G. Jin, X. Cao, G. Cong, C. S. Jensen, and X. Cheng, "Exploring progress in multivariate time series forecasting: Comprehensive benchmarking and heterogeneity analysis," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 1, pp. 291–305, 2025.
- [33] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto *et al.*, "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 3, 2021, pp. 747–769.
- [34] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [35] J. G. Wardrop, "Some theoretical aspects of road traffic research," *Proceedings of the Institution of Civil Engineers*, vol. 1, no. 3, pp. 325–362, 1952.
- [36] M. Beckmann, C. B. McGuire, and C. B. Winsten, *Studies in the Economics of Transportation*. Yale University Press, 1956.
- [37] Y. Sheffi, *Urban Transportation Networks: Equilibrium Analysis with Mathematical Programming Methods*. Prentice-Hall, 1985.
- [38] R. B. Dial, "A probabilistic multipath traffic assignment model which obviates path enumeration," *Transportation Research*, vol. 5, no. 2, pp. 83–111, 1971.